

LAB ASSIGNMENT-02

Exploring packet sniffer and packet Analyzer:

Question-01: What are the different protocols you observe at the following layers of the protocol stack?

Solution: (A) Application layer

- HTTP
- HTTPS/TLS
- DNS

(B) Transport layer

- TCP
- UDP

(C) Network layer

- IPv4

Question-2: What is the total amount of data being received for the following two cases?

(A) When you access <http://neverssl.com/>

(B) When you access <https://www.cornell.edu>

Solution:

(A) While accessing <http://neverssl.com/> the total data received are 2700 bytes.

(B) While accessing <https://www.cornell.edu> the total data received are 19,243 bytes.

Question-03: Answer the following when you access the site `http://neverssl.com/`.

(A) How many HTTP GET request do you observe? List down the GET request.

Solution: There are 3 HTTP GET requests

- GET /online HTTP/1.1
- GET /online HTTP/1.1
- GET /favicon.ico HTTP/1.1

(B) What version of HTTP the server is running.

Solution: HTTP/1.1

(c) For each of the HTTP GET request find out:

(i.) The total number of TCP segments being received

Solution: 3 TCP segments

(ii) The total amount of data being received in the corresponding HTTP response message.

Solution: 2273 bytes of data

(d) Identify TCP three way handshake used to establish the connection with the `http://neverssl.com` specify SYN, SYN-ACK, ACK packets and write the sequence numbers for each.

<u>Solution</u> :	<u>Packet</u>	<u>Sequence Numbers</u>
	SYN	950965917
	SYN-ACK	7225813718 (ACK: 515996719)
	ACK	515996719 (ACK: 7325813718)